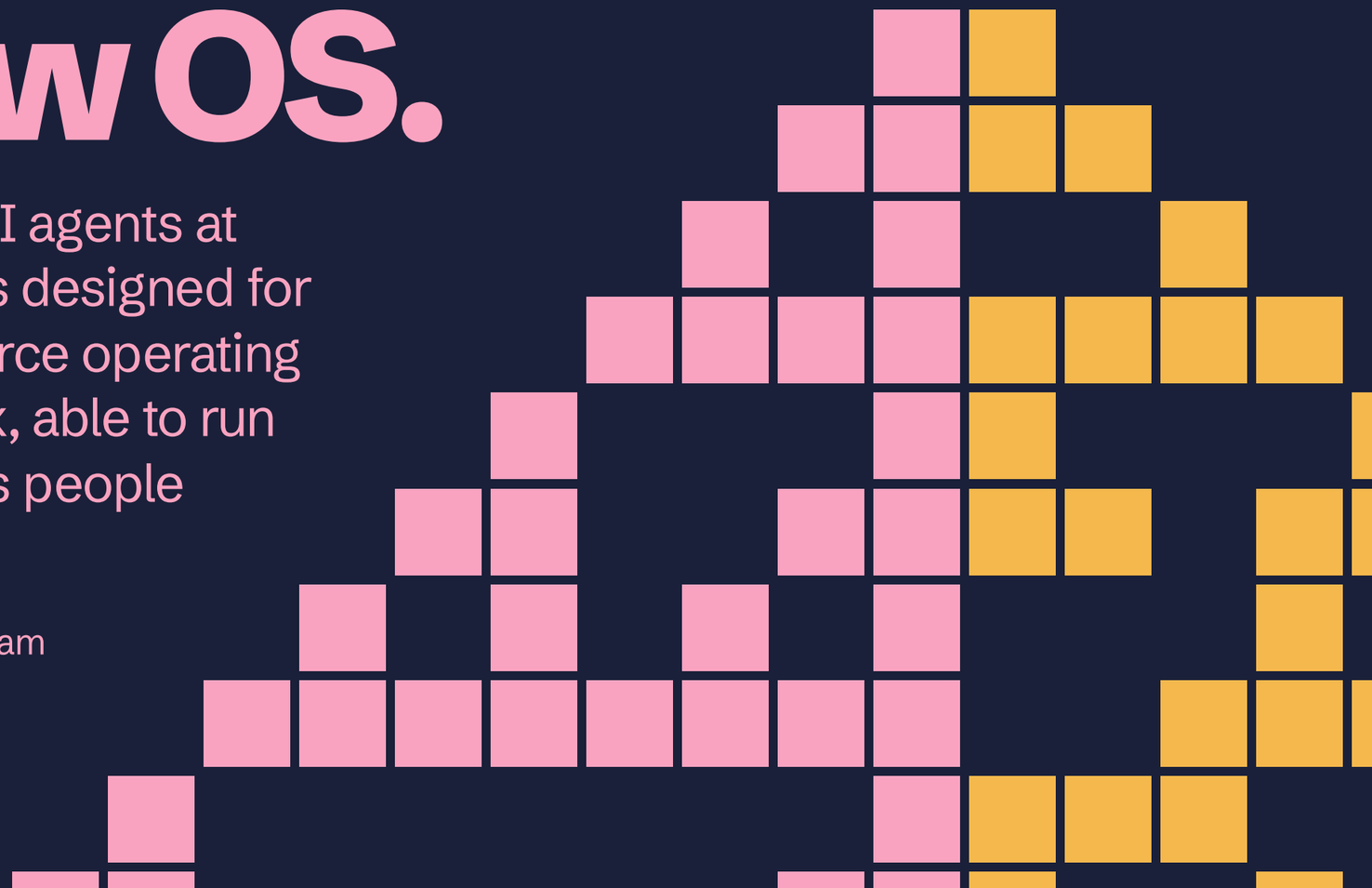
Automaton

A new user needs a new OS.

By 2028 there will be about 1.3 billion AI agents at work. Every computer they will use was designed for one person. Automaton is an open-source operating system rebuilt for them: based on Linux, able to run Windows applications, on the machines people already own.

Pre-seed round. I am Ronald Djakou, the founder, and I am looking for co-founders. rkouatch@caltech.edu



1 1.3 billion agents: IDC forecast, quoted by Microsoft, 2025.

The models are not the problem.

Two years ago the best model completed 12 % of everyday tasks on a real desktop. The best now completes 86 %, more than a person does. Yet the way the world works has hardly changed.

4 months

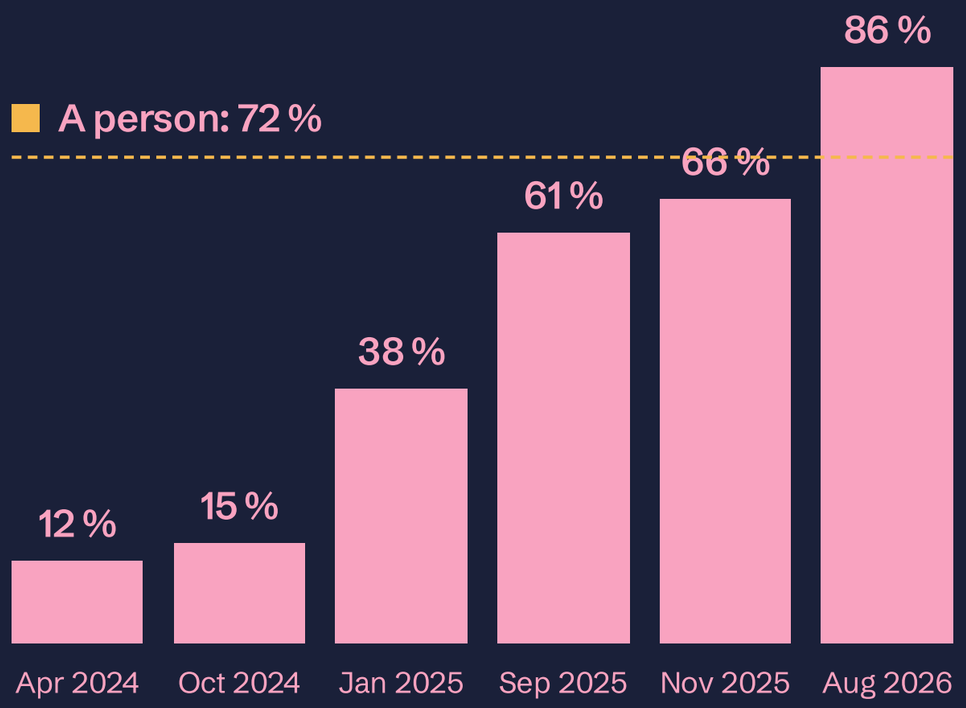
is all that separates open models from the best. Good enough for most everyday work.

10 %

of American firms use AI to produce what they sell.

Companies and investors think applications are what is missing. But every application runs on an operating system, and that is where the obstacle is.

Everyday tasks completed on a real desktop



Best published score of a general model on the OSWorld benchmark at each date. The benchmark was revised in July 2025, so the earlier and later figures are not strictly comparable.

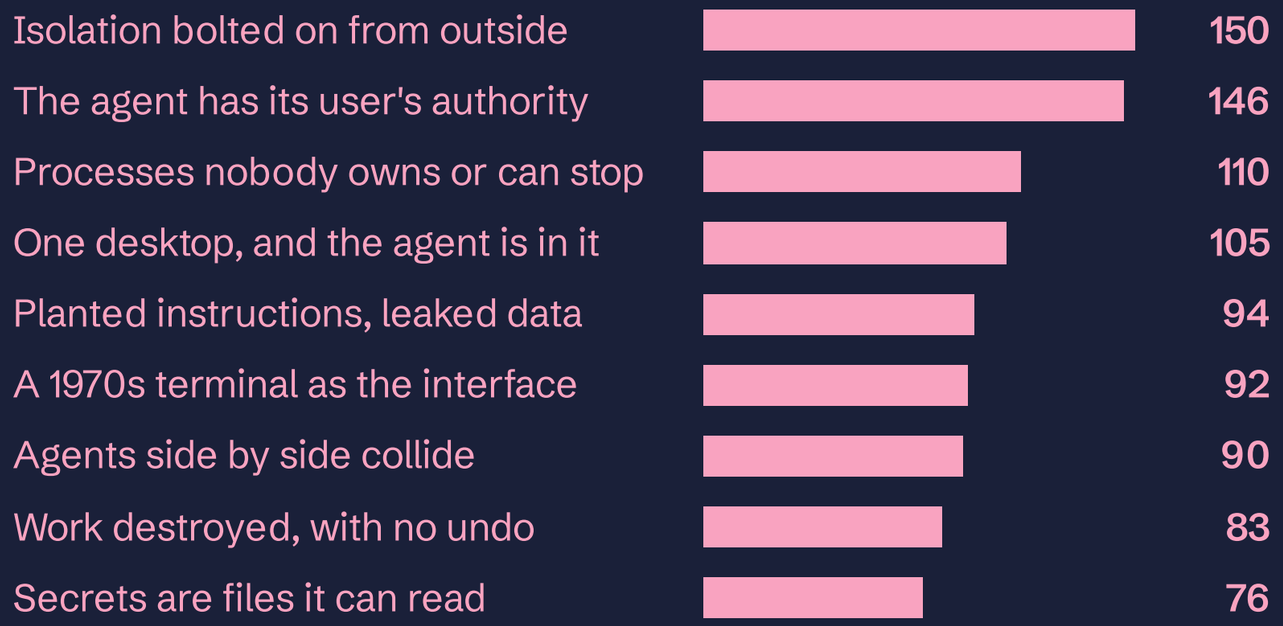
Sources: OSWorld benchmark leaderboard; Epoch AI, May 2026; US Census Bureau business survey, 2025.

Built when people were the only users.

Operating systems were designed for one kind of user: a person at a screen, with a keyboard and a mouse. AI agents can now operate software as well as people can, but the system has no place for them, so they borrow a person's screen, account and files.

I collected 1,214 public reports of agents failing on today's systems and sorted 1,168 of them by cause. Almost none blame the model. They blame the system around it. In my own test, a real agent read a planted secrets file and repeated it 17.4 seconds after it started. Nothing stopped it.

What goes wrong, by number of reports



My field study, September 2026: 1,214 reports from 721 pages, 1,168 of them sorted into twelve groups. The nine largest are shown.

Make the agent a user of the system.

Today an agent is a program that runs under your name, with your screen and your access. Automaton treats it as a second kind of user, with a place of its own.

	Today: an app that runs as you	Automaton: a user of the system	
Screen	Yours. One agent at a time	Its own desktop. You keep yours	**Works today**
Files	Written straight into your files	Its own layer. You keep or discard	**Works today**
Watching	One window, one agent	Every agent on one view, live	**Works today**
Identity	Your account, and all it can reach	Its own account, with what you grant	**Stage one**
Secrets	Any file you can read, it can read	Used through the system, never read	**Stage one**
When you leave	Stops when the laptop sleeps	Works on; you reach it from your phone	**Stage one**
Models	In someone else's cloud	Local or cloud, run by the system	**Stage one**
Tools	Whatever happens to be installed	From a store, reviewed for safety	**Later**

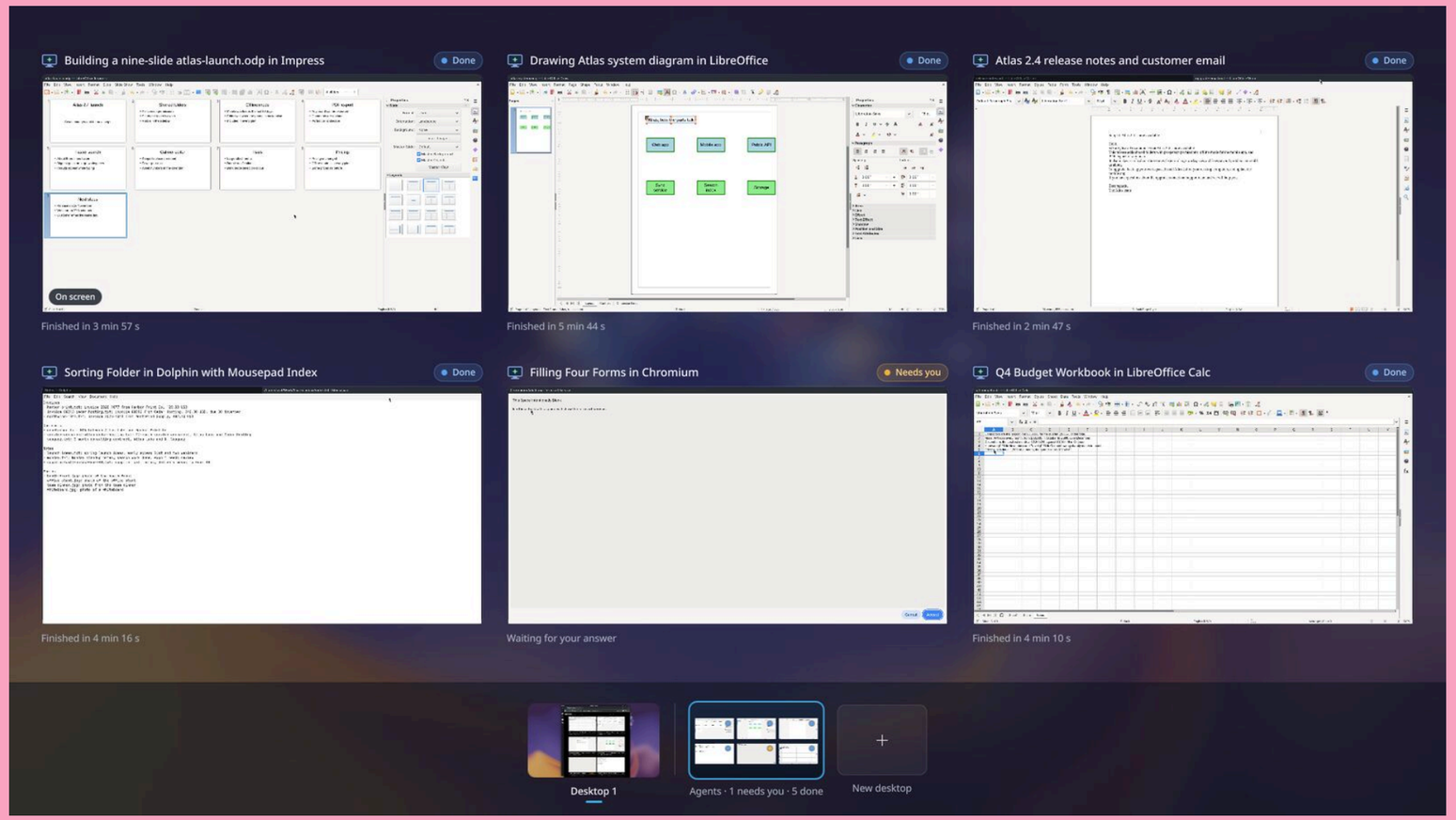
It already works on one ordinary PC.

On today's systems only one agent can use the computer at a time, and you cannot use it while it does.

- Here, six agents in six ordinary apps finished six long tasks in one run of 22 minutes, each on its own desktop.
- Their owner kept typing beside them: a key press reached the screen in 31 to 43 ms, against 24 to 26 ms with no agents.
- One more agent costs 221 MB of memory. Built in three days, on an 8 GB machine with no graphics card.

Watch it, three minutes:
youtu.be/Ugkh9cJ_mVg

Task View, 9 November 2026: every agent's desktop on one screen.



Where the revenue comes from.

The OS will be open source. It has to be, to gain any trust, and it is how the OS gets onto people's laptops. In the early days the revenue comes from companies: an enterprise edition that lets them use local and cloud models at full capacity without compromising their data.

$30

for each person, each month

A 500-person firm, three agents each: the monthly bill



Cloud PCs that stay on	**$438,000**
Cloud PCs, working hours only	**$105,600**
Automaton, on PCs the firm owns	**$15,000**

Cloud PC for an agent at Microsoft's $0.40 an hour; working hours are 8 a day, 22 days. Model use costs the same on both sides. The firm's PCs stay on while agents work; their power and upkeep are not counted.

Now: the enterprise edition, agents included

Later: a fee on every sale in the stores

Later: a fee for reviewing what enters the stores

Later: licences to hardware makers

Source: Microsoft's price for Windows 365 for Agents, 2026.

Ambitious enough to look unrealistic.

AI now writes most software, so a small product can be copied in a weekend. An operating system cannot. It takes compatibility with the applications people already use, a security record, and the trust of the people who rely on it. On top of it come the stores that every tool and model for agents plugs into. That is why the plan is this large.

Stage one

What this round pays for. Built on Linux, for laptops and desktops. Windows applications run on it.

Pays: enterprises.

Stage two

Phones, watches and headsets.

The stores

Tools for agents; models tested for safety.

Pays: a fee on each sale.

Hardware

Designed from what we learn.

Pays: licences to makers.

Our models

If the data says so.

Why the incumbents will not.

$123 billion

Their revenue depends on today's design

Apple's services earned that last year, a quarter of the company, much of it from apps sold to people. A system where agents do the work cuts into it. Apple and Google cannot give that up while a new income grows.

1.6 billion

They cannot break what a billion people use

That many devices run Windows, and every change has to leave them all working. So anything new arrives slowly and optional: Microsoft announced Recall, withdrew it within four weeks, and shipped it eleven months later, switched off by default.

430 replies

Their users do not trust them with it

Nearly all hostile, when Microsoft said Windows was becoming an "agentic OS". It later dropped the phrase. "I just want my computer to be a computer." This is why the OS has to be open source: anyone can read what it does.

Microsoft's own attempt, Agent Workspace, has been a preview since 2025: switched off by default, on Windows only, with no view of several agents at once and no undo of what an agent changed. A new company can build the whole system around the new user.

Sources: Apple, results for fiscal 2026; Microsoft, annual report 2026 and announcements; my field study.

Built so far.

I built all of this before any funding, working with AI agents.

3 days

from the first line of code to six agents at work on one PC, in 200 commits.

1,214

reports of agents failing on today's systems, audited, and 11 controlled experiments. Published: kdronald3.github.io/aios-agent-os-pain

697

Windows applications, drawn at random, tested on the Linux base with Wine and my own fixes. 46 % already run, 66 % at least in part.

13.6 s

a task with my computer use, against 20.8 s from screenshots. Faster with all three models tested, in 456 runs.

Who is building it

I am Ronald Djakou. I study mathematics at Caltech and do research in quantum systems and cellular automata. IBM awarded me Quantum Excellence at its Qiskit school in 2024. I built the simulation tools my research groups lacked, and cut one analysis from 59 GB of memory to 2.5 GB.

I am looking for co-founders, then engineers.

I am raising $2 million.

There are no users yet, and I am the only founder. This pre-seed round pays for a team of five (me, one or two co-founders, then engineers) and takes Automaton from a working demo to a system its first pilot customers use every day, in 18 months.

Month 3

The first co-founder or engineer has joined. The demo is hardened.

Month 6

The seat layer on Linux, version one. Security review started.

Month 9

The team is five. Local models drive agents, measured.

Month 12

Automaton installs as its own system. The first pilot pays.

Month 18

The pilots' Windows applications run. Ready for a seed round.

Where it goes: 58 % pay for five people (my pay at $100,000 a year; four more at about $240,000 with costs, joining over months 3 to 7), 4 % finding them, 15 % AI compute, 6 % security and legal, 5 % hardware, 4 % pilots, 8 % reserve. **Contact:** Ronald Djakou, rkouatch@caltech.edu. **Demo:** youtu.be/Ugkh9cJ_mVg